An Agent-Based Simulation Model for Testing Novel Obesity Interventions in School Environment Design

Jeri Brittin, Ozgur M. Araz, Adrian Ramirez-Nafarrate, and Terry T.-K. Huang

***Abstract*—Studying the impact of built-environment changes on human behavior and health is empirically challenging. To overcome this challenge, there has been increasing recognition of the usefulness of systems' approaches for scenario testing to understand and prioritize such public health interventions. Limited research has shown a potential effect on the positive energy expenditure as a result of the use of dynamic furniture that promotes micromovement among school children. In this article, we aim to determine, via experiments in simulated populations of elementary school students, whether the use of dynamic furniture in the school environment has an impact on the childhood obesity over time. An agent-based model is developed to simulate three physical activity (PA) profiles over a period of six years with and without the dynamic furniture. Based on the prevalence trends from the experiments, there is no apparent impact of dynamic furniture use among boys, regardless of activity profile category. However, there are some pieces of evidence of differing trends among girls with a low PA profile. Although the intervention produces a marginal change in obesity prevalence in one population, the dynamic furniture is not a sole solution to the obesity epidemic. However, the use of dynamic furniture in schools may be a worthwhile component of prevention interventions.**

***Index Terms*—Agent-based modeling (ABM), built environment, childhood obesity, decision analytics.**

I. Introduction

IN THE United States, the prevalence of childhood obesity began to rise in the early 1980s and tripled by the year 2000 [9]. Close to one-fifth of the population of America, school children are now obese, and this trend has stayed persistently high over the past decade [25]. A large body of literature has shown significant associations between children's physical activity (PA) levels and obesity and related diseases [32]. In recent years, research on childhood obesity has been increasingly focused on environmental correlates of obesity-related behaviors, such as PA, as individually focused interventions have been both difficult to sustain and have had relatively little positive population-level impact [19], [39]. Notably, the design of the school environment has been an intervention target of substantial interest [13], [21], [24], [28]. A 2012 Institute of Medicine (now called the National Academy of Medicine) report pointed out that "[c]hildren spend up to half their waking hours in school. In an increasingly sedentary world, schools, therefore, provide the best opportunity for a population-based approach for increasing PA among the nation's youth" ([31], p. 333). The percentage of school children spending PA times during a day has been steadily decreasing the United States (U.S.) [32] and recent findings indicate that the majority of children and youth in the U.S. do not meet PA recommendations (78.4%) [37]. Since the physical educational programs have been under funded in many school districts, innovative approaches to increase the children's activity levels would have positive public health impact.

In this article, we develop and use a complex systems simulation modeling approach to assess the impact of these innovative interventions and inform public health policy making. An agent-based modeling (ABM) is developed and simulated, and the results are analyzed for statistical significance of the novel design-thinking-based intervention. The relevance in the problem under study is remarkable because it allows policy makers to avoid high implementation costs of ineffective policies that could be evaluated only in the long term. Although ABM has been applied to multiple fields, including architecture (e.g., the evaluation of evacuation plans from buildings, urban design, etc.) and design of intervention strategies to promote PAs in children, to the best of our knowledge, studies that combine both subjects are scarce. Therefore, this article contributes to the existing literature with a novel application of ABM that could allow policy makers on the design of effective intervention plans. Next, we review the literature in two main related fields: first, we review the status and significance of the childhood obesity problem with novel intervention studies, and then review the application of the complex systems approach with ABM models studying childhood obesity.

II. Literature Review

The childhood obesity epidemic is continuing to be a major public health problem in the United States as the prevalence reached 19.3% for children and adolescents in 2017–2018 [20].

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Jeri Brittin is with the HDR, Inc., Omaha, NE 68106-2973 USA (e-mail: jeri.brittin@hdrinc.com).

Ozgur M. Araz is with the College of Business, University of Nebraska-Lincoln, Lincoln, NE 68588 USA (e-mail: oaraz2@unl.edu).

Adrian Ramirez-Nafarrate is with the Facultad de Ciencias Económicas y Empresariales, Universidad Panamericana, Guadalajara 45010, Mexico (e-mail: aramirezn@up.edu.mx).

Terry T.-K. Huang is with the Health Policy Department, Center for Systems and Community Design, City University of New York Graduate School of Public Health and Health Policy, New York, NY 10027 USA and NYU-CUNY Prevention Research Center, City University of New York Graduate School of Public Health and Health Policy, New York, NY 10027 USA (e-mail: terry.huang@sph.cuny.edu).

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Obesity has been considered as a complex systems problem by public health organizations globally [6], [38], and factors that influence the food consumption behavior of individuals and food consumption trends in the population are well studied in the literature. However, few studies have used simulation modeling to assess physical-activity-based interventions. Here, we present a review on the childhood obesity status and policies in the most recent years, which is followed with a review on the use of ABM approach.

A. Review on Childhood Obesity

First, there are several environmental factors that significantly affect food accessibility and intake, and these studies in the literature show the evidence for the population-level trends. On the other hand, there are behavioral factors that can have great impact on the food consumption behavior of individuals, one of which can be listed as the nutritional information seeking behavior [30], [58]. Therefore, educational interventions can be effective through behavior change as well. Finally, social factors and peer group can also influence what individuals are consuming [2], [6], [18]. Jódar *et al*. [35] presented a model of a social transmission of obesity for infants but did not consider any variation on behavioral or social factors that can cause the disease progression. They concluded that more than the environmental factors, social networks can play a key role in understating the obesity epidemic evolution. Our article falls into a category that investigates the peer and built-environment factors in childhood obesity epidemic using a computational modeling approach.

Furniture is an important component of the built environment in schools that can be easily modified, given funding availability. The school-based studies have found that children using stand-biased desks expended significantly more energy than those using standard chair-height desks [7], [8], [10]. In addition, it has been argued that furniture that allows and encourages children to move increases PA levels and improves learning outcomes, and that children's ergonomic needs include chairs with movable seats and chair–table combinations that are adjustable in height. A lab-based study using accelerometers to measure PA showed that children using "active" furniture that facilitated natural movements and fidgeting had significantly higher average activity counts per minute than children using the traditional school furniture (mean activity counts per minute were 40.82 versus 9.81 on active versus the traditional furniture, respectively, $p < 0.05$) [22]. Fig. 1 shows an example of a classroom outfitted with dynamic and mobile furnishings.

Nonexercise activity thermogenesis (NEAT) is defined as energy expended that is "not from sleeping, eating, or sports-like exercise" but includes light-intensity PA, such as fidgeting, typing, sitting, talking, standing, and nonexercise walking [44]. The use of dynamic furniture in the school environment may promote NEAT. Research has shown that increases in NEAT can impact the overall energy balance and can provide protection against fat gain and obesity [42], [45], [63]. Levine *et al*. [43] suggested that the obesity epidemic may be related to what they call "chair-enticing environments," and they recommended changing school, home, and work environments to encourage more active living. However, given limited data on built-environment strategies to promote NEAT, more research articles are needed to recommend strategies in the school environment.

Fig. 1. Dynamic furniture in a modular classroom arrangement. Chair roll, tip, rock, swivel, and accommodate forward- or backward-facing sitting positions, and the heights of desks and seating are adjustable. (Photo: VS).

B. Review on Systems Simulation and ABM

Built-environment interventions are difficult to study empirically due to the constraints of time, money, and sampling issues. As a result, virtual testing using systems models has gained popularity in recent years. Agent-based modeling (ABM) is a rule-based modeling approach in which each agent (i.e., autonomous, decision-making actor) in the system is represented in computer code. Agents are placed in the context of a dynamic, adaptive environment with rules governing their decisions and actions that give rise to behaviors of the whole system. The agent-based models are stochastic in nature and can, thereby, represent natural variations found in the real world, making it useful to address the real-world problems and inform intervention design [11]. The examples of ABM in the article of obesity include a model by Zhang *et al*. [66], which analyzed the effects of social ties on obesity and related stigma. In addition, in the context of school health, Fetta *et al*. [17] presented an agent-based model using social networks to inform smoking prevention interventions and Sun *et al*. [60] use the online social network sensors to estimate local obesity outcomes.

As obesity is a serious problem in the U.S. children and adolescent population [50], Auchincloss [4] performed a study via agent-based simulation modeling to explore patterns that generate income disparities in diet and the relation with obesity. Social group and peer influence are not only observed on food consumption but in other consumer behaviors as well, including fashions [33] and technology products [34]. These types of behaviors are modeled and simulated through one of the computational methods from physics literature called the opinion dynamics that studies the evolution of complex systems and the

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behaviors forming these systems ([23], [36], [47], [53], Sun and Muller, 2012). Hall *et al*. [26] also presented a mathematical model of metabolic dynamics for children, which accounts for healthy growth. The model, we used in this article, is in agent-based simulation framework and we describe the details in Section III. Giabbanelli *et al*. [23] presented a model of obesity in which the food intake and PA of individuals are modeled based on the social influences of the network that they are in. Using an individual-level modeling approach with social ties is taken into account.

In this article, we present an agent-based simulation model to examine the impact of the built environment on overweight and obesity in elementary school children, considering empirically derived dynamics of children's energy balance and growth. Specifically, we sought to understand whether exposure to "active" furniture in the school environment would significantly affect students' energy balance over time and lead to a reduction in the prevalence of childhood overweight and obesity. The aim of the research is informing the design process using dynamic furniture in school environment and assess the impact before implementation. Thus, the complex systems framework and the ABM proposed in this article would enhance the decision-making process to design effective the obesity prevention policies that would consider the build environment as a part of the intervention design program for school-aged children. This novel application of ABM not only helps to understand the effect of dynamic furniture on childhood obesity but also provides with more information and insight to decision and policy makers about this issue before expending significant amount of money in the redesign of classrooms and evaluating its impact in the long term.

III. Agent-Based Simulation Model

Due to the nature of the intervention that we want to test and evaluate for informing public health policy, we had to take a bottom–up approach, which is best reflected in an ABM. Individuals constituting a special population in the article have unique attributes, e.g., gender, weight, height, etc., and they interact with each other and with the environment dynamically; the ABM is a multiagent model (Durand *et al*., 2012). This type of models has been widely used recently for evaluating public health policies [40], [54], [58].

A. Base Model

We propose an agent-based simulation approach to incorporate the interrelations and complexity of the system under study. An agent-based simulation is a model to assess decision makers and they include autonomous agents characterized with a set of attributes and rules that determine their behavior and, in consequence, the evolution of the system [41].

Given that the obesity is a problem with characteristics of complex adaptive systems, ABM is an approach that could be used to analyze noncommunicable diseases, such as obesity [49]. Hennessy *et al*. [29] acknowledge that ABM can be useful to study obesity interventions for child obesity. In their work, they sketch the model framework to study the effect of food consumption and PA on child obesity. However, they did not complete the construction and analysis of the model.

The framework presented in this article is based on an agent-based simulation proposed by Ramirez-Nafarrate and Gutierrez-Garcia [57] to analyze the prevalence of childhood obesity. That model provides a foundation upon which to incorporate and test the intervention of interest. In this model, the agents are children with the following attributes: age, gender, weight, height, body mass index (BMI), weight status category (underweight, healthy weight, overweight, and obese), calorie intake, and energy expenditure. We have adopted the ABM model to be reflective of the children's PA in the school environment to answer the research question of how school environments can be designed in a way to help avoid rising prevalence of obesity and overweight. The research question with the modeling environment is conceptualized in Fig. 2. The aim of the research is informing the design process using dynamic furniture in a school environment and assess the impact before implementation. Thus, the ABM model proposed in this article would enhance the decision-making process to design efficient and effective intervention policies, in particular, regarding the type of furniture that fosters PA in elementary school. The analytics models are tools that could be helpful to make decisions on public health issues [1], [3], [5].

The model allows one to track the weight and height of a child population throughout the elementary school years. Therefore, when the simulation run starts, age, weight, and height are drawn from a random probability distribution for a predetermined population size of children of the same gender. Height and weight are proportional to age. Then, BMI is obtained for each agent to determine his or her weight status based on the Centers for Disease Control (CDC) and Prevention's growth charts [14]. These charts define the weight category by gender using percentiles of the BMI (<5th percentile is underweight, 5th–<85th percentile is normal weight, 85th–<95th percentile is overweight, and ≥95th percentile is obese). BMI percentile values change nonlinearly for children in elementary school age.

In each day of the simulation run and for each agent, height is updated, assuming a linear growth trend of children in elementary school age. However, weight is updated based on a function that includes the updated height, calorie intake, and calorie expenditure. Calorie intake is randomly determined around a minimum amount required for children at a given weight. The overall energy expenditure is determined based on the energy balance equation of Cutler *et al*. [16] and Schofield *et al*. [59], as follows:

$$\mathrm{kcal}_{\mathrm{out}} = ((\alpha + (\beta \times W) \times 1.71) + (E \times W) + 0.1\,\mathrm{kcal}_{\mathrm{in}} \tag{1}$$

where

$\mathrm{kcal}_{\mathrm{out}}$ = energy expenditure in kilocalories;

W = weight in kg;

$\mathrm{kcal}_{\mathrm{in}}$ = energy intake in kilocalories (assumes 95%–117% of minimum required);

E = energy expenditure in kilocalories due to PA;

α and β = constants based on gender [16].

1.71 is a multiplier that adjusts for higher average basal metabolic rates in children versus adults [59].

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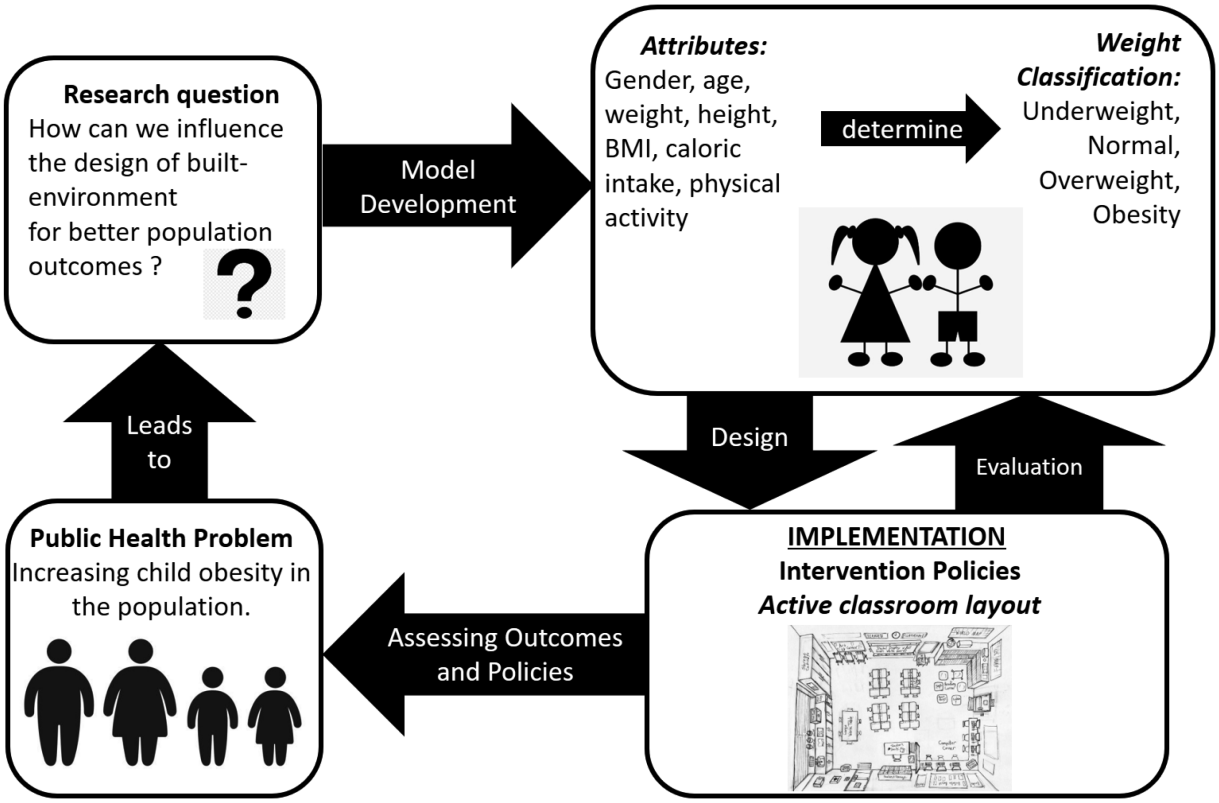

Fig. 2. Research conceptualization and design.

Estimates of E are determined based on studies from Harrell *et al*. [27] and three profiles of PA are defined: low PA profile with energy expenditure range of 0.5–3.5 kcal/kg, medium PA profile with energy expenditure range of 1.5–4.5 kcal/kg, and high PA profile with energy expenditure range of 2.5–5.5 kcal/kg.

The updated weight is obtained through a function that considers the increase in height and the difference between calorie intake and expenditure. With the updated weight, BMI and weight category are also updated. The simulation runs for a predetermined number of years (e.g., number of years in elementary school) and overweight and obesity prevalence is analyzed based on the percentage of overweight or obese children of the same gender. The base model's simulation algorithm is shown in Fig. 3.

B. Model Validation

The experiments presented in Ramirez-Nafarrate and Gutierrez-Garcia [57] used the model described above to track the prevalence of obesity of children aged 6–12 years. In that article, the initial state of the agents was given by the expected distribution of weight and height of boys and girls aged 6 based on the CDC growth chart. Then, as the time evolves in the simulation model, the weight and height of the agents are updated based on the assumptions presented in Section III-A depending on the caloric intake and PA profile. The results included 95% confidence intervals for the percentage of boys in each weight category at the end of the simulation run, depending on the PA profile. These results are given in Table I. According to these results, for boys with medium level of PA, the expected percentage of underweight boys at age 12 is between 0% and 0.32%, the expected percentage of normal weight at that same age is between 79% and 84%, while the expected percentage of obese and overweight boys is between 15% and 21%. The CDC growth charts suggest that 5% of children are expected to be underweight at the age of 12. Similarly, 80% are expected to have healthy weight, and 15% are expected to be overweight or obese based on the reference population. Hence, using a medium PA profile, the results show that the base model underestimates the proportion of underweight children, while children with healthy weight and overweight/obese are within the expected proportion. Therefore, as the objective of this article is to track the overweight/obese prevalence in children by means of intervention in school furniture, we believe that the base model is valid to evaluate the effectiveness of this intervention incorporating the changes described in Section III-C.

C. Intervention Model

The "active" furniture intervention in the school environment allows for changes in the energy expenditure portion of the energy balance equation. The active furniture use in schools can provide continuous and sustained activity for children to enable energy expenditure while being cost-effective compared with other targeted programs. Therefore, they have been considered in new school design programs; thus, we considered this as an

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```
Agent-based simulation for obesity tracking without intervention
Set lastday = total simulated days
Set PAP = Physical Activity Profile
Generate population size N of agents
For i = 0 to N
    Set initial values of age, weight and height for agent i
    Compute BMI and weight status category for agent i
EndFor
For day = 1 to lastday
    For i = 0 to N
        Set caloric intake for agent i
        Set energy expenditure for agent i based on PAP
        Update weight and height for agent i
        Compute BMI and weight status category for agent i
        If (day mod 365 = 0) Then
                Increase age of agent i by 1
        EndIf
    EndFor
    Compute population-level statistics
EndFor
```

Fig. 3. Base model simulation steps.

TABLE I
95% CONFIDENCE INTERVALS OF THE PERCENTAGE OF BOYS IN EACH WEIGHT STATUS CATEGORY AT THE END OF THE SIMULATION (AGENTS AGED 12) FOR EACH INTENSITY LEVEL OF PA

Weight Category	Low PA (%)	Medium PA (%)	High PA (%)
Underweight	(0, 0)	(0, 0.32)	(12, 18)
Normal Weight	(16, 20)	(79, 84)	(81, 87)
Obese & Overweight	(79, 83)	(15, 21)	(0, 0.50)

intervention to be tested using simulation modeling. In the model, the intervention is assumed to have no impact on children's calorie intake levels. Several studies have developed formulas to convert accelerometer-measured activity counts to energy expenditure, and a comparative study by Trost *et al.* [64] determined that a formula from the article presented in [55] was most accurate in predicting energy expenditure from accelerometer measures performed during light-intensity PA, such as walking.

Therefore, the intervention model uses the formula of Puyau *et al.* [55] to convert activity counts to energy expenditure

$$\mathrm{AEE} = 0.0183 + 0.00001 \times \mathrm{AC} \tag{2}$$

where

AEE = average energy expenditure in kcal/(kg·min);

AC = activity counts per minute = 40.82 counts/min for active furniture or 9.81 counts/min for the traditional furniture [22].

Based on the knowledge of a typical school schedule, the model assumes that elementary-school-aged children spend an average of 2.4 h/day (or 144 min/day), over the course of a year, using the dynamic school furniture [16]. This assumption represents 876 h at school in a year. This number is consistent with the number of hours per year spent at school in the United States, which is between 700 and 1000 h, approximately depending on the grade and the state (National Center for Education Statistics.[1]

This is based on a hypothetical intervention whereby dynamic furniture is used throughout the school. The intervention model simulation steps are shown in Fig. 4. Energy expenditure in the intervention model includes the calculations from the base model that uses a random function from distributions based on the children's overall PA profiles, along with a calculation of daily energy expended while using dynamic school furniture.

[1][Online]. Available: https://nces.ed.gov/programs/statereform/tab5_14.asp

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```
Agent-based simulation for obesity tracking with intervention
Set lastday = total simulated days
Generate population size N of agents
For i = 0 to N
    Set initial values of age, weight and height for agent i
    Compute BMI and weight status category for agent i
EndFor
For day = 1 to lastday
    For i = 0 to N
    Set caloric intake for agent i
    Set energy expenditure for agent i based on PAP
    If (Intervention = Yes) Then
            Energy expenditure for agent i = Energy expenditure for agent i
                                             + 144 * (0.0183 + 0.00001 * 40.82)
    Else
            Energy expenditure for agent i = Energy expenditure for agent i
                                             + 144 * (0.0183 + 0.00001 * 9.81)
    EndIf
    Update weight and height for agent i
    Compute BMI and weight status category for agent i
    If (day mod 365 = 0) Then
            Increase age of agent i by 1
    EndIf
  EndFor
  Compute population-level statistics
EndFor
```

Fig. 4. Intervention model simulation steps.

TABLE II
AGENT-BASED MODEL INPUTS AND OUTPUTS

Model Inputs	Value	Source	Model Outputs	
$\boldsymbol{H_1}$	height on day 1 = random selection of height in cm from normal distribution based on average per age where $N(\mu,\sigma)$ = $N(115.66,5)$ for males and $N(115.01,5)$ for females	CDC 2010	$\boldsymbol{H_t}$	$\boldsymbol{Weight\ Status\ Category_t}$ Obese, Overweight, Normal Weight, Underweight, calculated from H_t and W_t based on age and gender
$\boldsymbol{W_1}$	weight on day 1 = random selection of weight in kg from normal distribution based on average per age where $N(\mu,\sigma)$ = $N(21,5)$ for males and $N(20,5)$ for females	CDC 2010	$\boldsymbol{W_t}$	

This new portion of the calculation is based on the conversion formula of Puyau *et al*. [55] and the activity count constants from the formula of Garcia *et al.* [22].

In Table II, a summary of the inputs and outputs in the simulation model is shown. Formulas for model parameters are given in Table III.

D. Experimental Setting

Six experimental scenarios (see Fig. 5) were defined for each child's gender based on the three (low, medium, and high) categories of PA profiles among children with two environments (i.e., intervention with active furniture in the school versus baseline with the traditional furniture in the school). Two sets of outputs were obtained for each simulated scenario, one for boys and another for girls. Like the base model, the intervention model converted raw BMI to BMI percentiles to assign BMI categories (i.e., underweight, normal weight, overweight, or obese) over time for each agent. The prevalence patterns of overweight and obesity were then observed over the elementary school years for boys and girls.

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TABLE III
AGENT-BASED MODEL PARAMETERS AND EQUATIONS

Parameter	Formula	Units	Components	Sources
AEE	$= 0.0183 + 0.00001 * AC$	Activity energy expenditure in Kcal per kg per minute	AC = accelerometer-measured activity counts per minute	Puyau et al. 2002, Taylor et al. 2020 Trost et al. 2006
$K_{out-child}$	$= \alpha + 1.71\beta W + EW + 0.1K_{in}$	child energy expenditure in Kcal per day	α and β = constants by gender: α_{female} = 829 β_{female} = 8.7 α_{male} = 879 β_{male} = 11.6 W = weight in kg K_{in} = energy intake in Kcal per day E = physical activity energy expenditure in Kcal per day 1.71 = constant multiplier accounting for children's greater base metabolic rate as compared to adults	Cutler et al. 2003 Schofield et al. 1985
E	$= 0.9\left(U_{low}(0.5, 3.5)\right) + 144(0.0183 + 0.00001 * AC)$ $= 0.9(U_{med}(1.5, 4.5)) + 144(0.0183 + 0.00001AC)$ $= 0.9\left(U_{high}(2.5, 5.5)\right) + 144(0.0183 + 0.00001AC)$	physical activity energy expenditure in Kcal per kg per day	U = random selection from uniform distributions of low, medium, and high physical activity energy expenditure ranges 0.9 represents 90% of day not spent on furniture 144 = minutes per day on school furniture (10% of total day) AC = accelerometer-measured activity counts per minute = 40.82 for dynamic furniture or 9.81 for traditional static furniture	Harrell et al. 2005 Puyau et al. 2002 Trost et al. 2006 Garcia et al. 2016
BMI_t	$= \frac{W_t}{H_t^2}$	body mass index on day t	t = day counter	
$BMI\ Percentile_t$	Conversion to percentiles based on age	percentile on day t		CDC Growth Charts
$BMI\ Category_t$	Conversion to weight status category based on percentile	category on day t	Underweight < 0.05 Obese >= .95 Overweight >=.85 and <.95 Normal weight >=.05 and <.85	

For each scenario, 1000 independent student agents were generated (50% boys and 50% girls) and population weight status trends were simulated for which the average values and standard deviations of the variables of interests were recorded. The model was developed using Netlogo 6.1.1 agent-based simulation environment [65]. Each simulation generated weight status categories based on the computed BMIs, and population prevalence outcomes for boys and girls at daily intervals are presented over a period of six years. The daily prevalence rates were reported for each gender group and initial prevalence values were set to 28.8% for boys and 29.7% for girls [48]. The trends of overweight and obesity prevalence were then graphed and compared based upon PA profiles and intervention conditions. The data were not intended to represent actual overweight and obesity trends that are occurring in a given population but rather to offer an opportunity to compare trends over time from a consistent starting point in two very specific environmental scenarios.

IV. RESULTS

The mean prevalence of overweight and obesity in the baseline model was different for males and females. For both genders, the intervention showed some effect of reducing the BMI for all activity groups, although the effect was stronger for the low activity group in both genders (see Table IV).

Fig. 6 shows the model-generated prevalence rates of weight categories in boys over six years, given the furniture intervention. The prevalence of obesity started above 30% and decreased markedly over time, reaching close to 0%, in both the control and intervention scenarios. Overweight prevalence at baseline was less than 10%, increased to nearly 20%, and then decreased markedly, again reaching close to 0%, in both the control and intervention scenarios, except for the low activity group. Total obese and overweight prevalence started above 40% at baseline and decreased to 0% during the elementary school years for both scenarios. Among females, obesity and overweight prevalence

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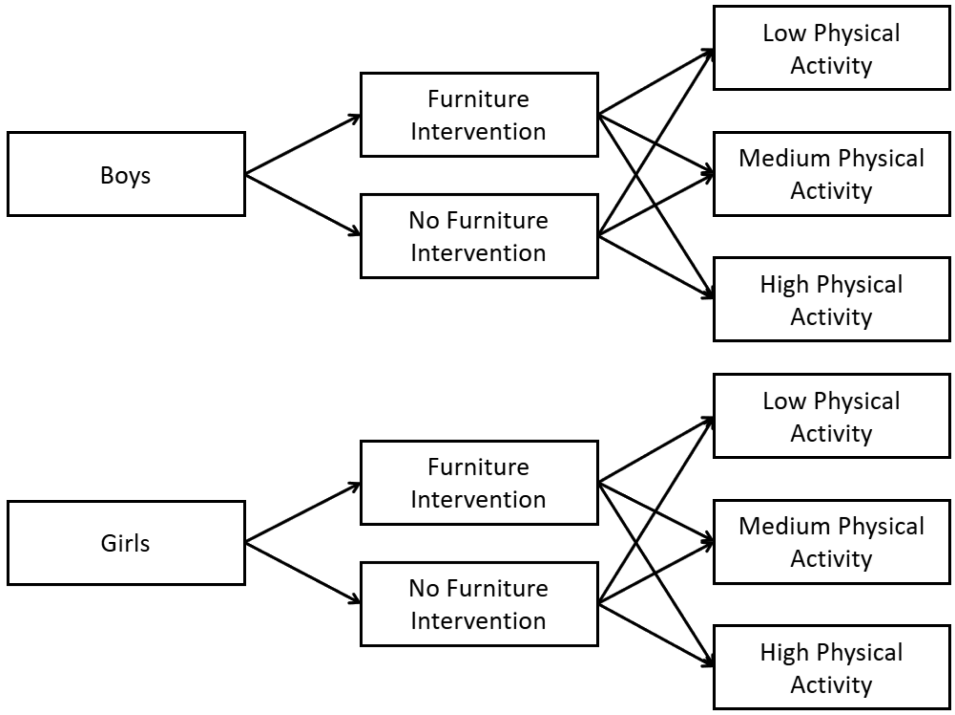

Fig. 5. Diagram of designed experiments.

TABLE IV
MEAN BMI AFTER SIX YEARS FOR BOTH CONTROL AND INTERVENTION GROUPS (STANDARD DEVIATIONS IN PARENTHESES)

Activity Levels	Males		Females	
	Control (Baseline)	Intervention	Control (Baseline)	Intervention
High	17.06 (1.50)	17.03 (1.49)	15.18 (1.11)	15.15 (1.10)
Medium	19.83 (1.83)	19.80 (1.83)	17.63 (1.30)	17.60 (1.31)
Low	23.22 (2.27)	23.17 (2.28)	20.64 (1.55)	20.60 (1.55)

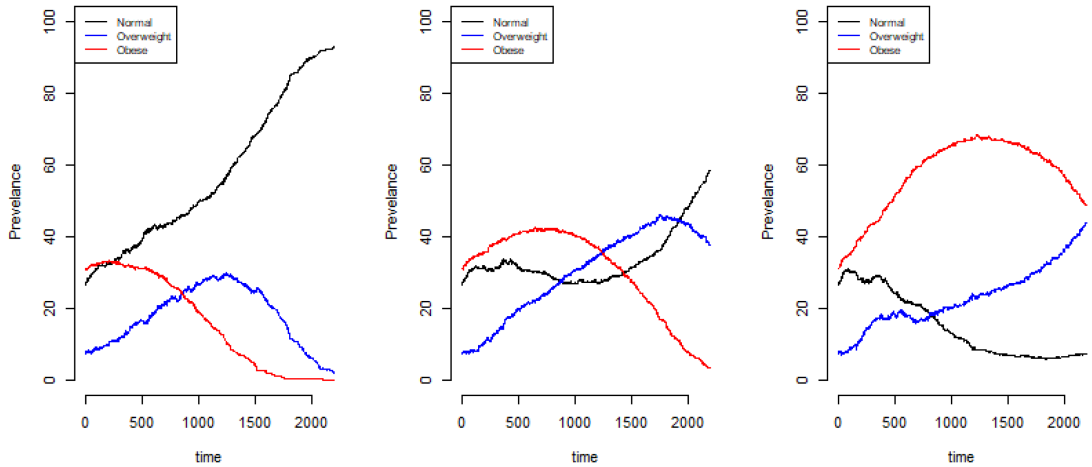

Fig. 6. Boys' weight status over time based on the activity levels when intervention is in place. Activity levels: High (left figure), medium (center figure), and low (right figure). Weight status: Normal (black line), overweight (blue line), and obese (red line).

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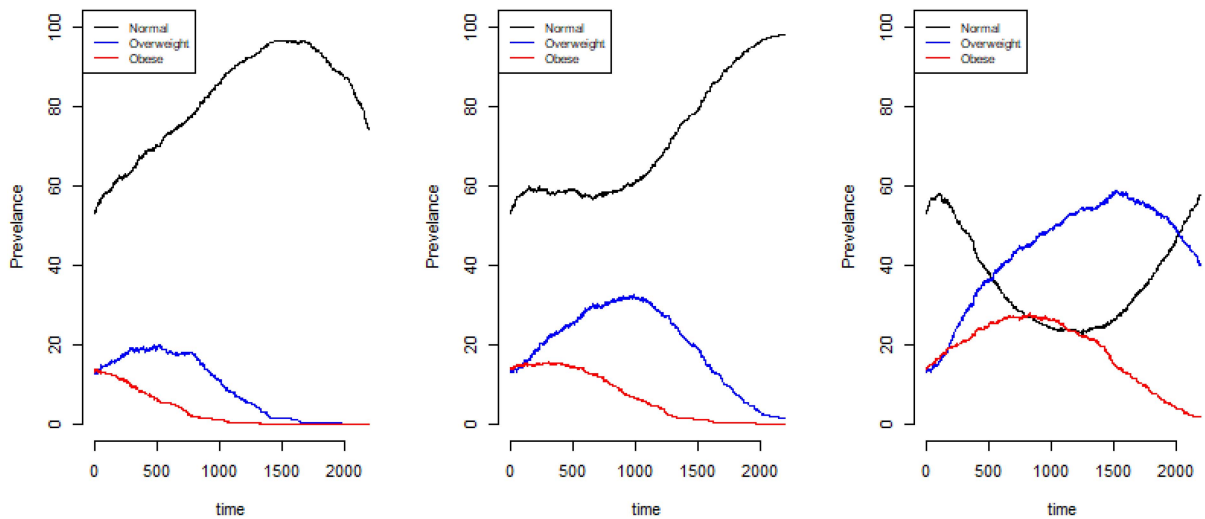

Fig. 7. Girls' weight status over time based on the activity levels when intervention is in place. Activity levels: High (left figure), medium (center figure), and low (right figure). Weight status: Normal (black line), overweight (blue line), and obese (red line).

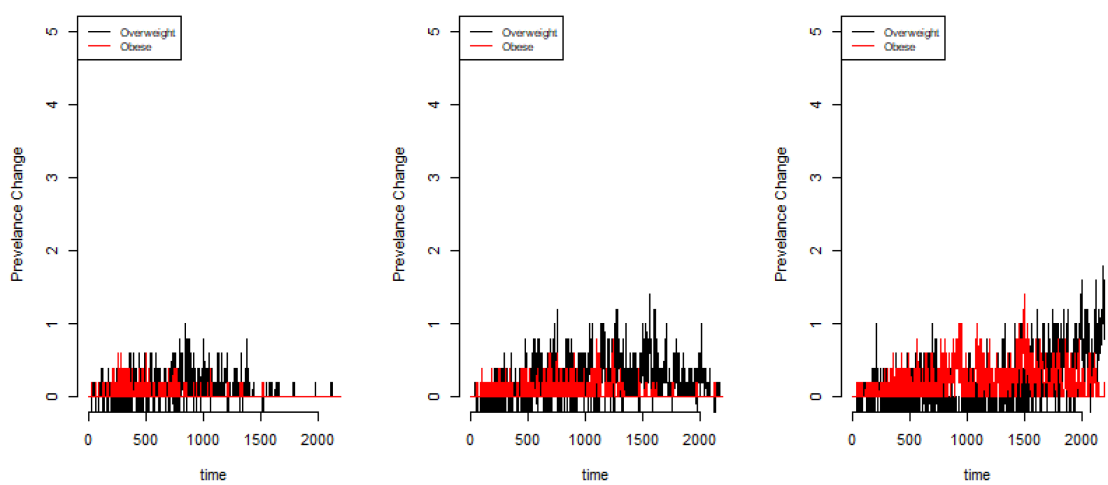

Fig. 8. Changes on boys overweight and obesity prevalence (%) (High, medium, and low activity levels, respectively) with the intervention in place.

patterns were quite similar for the intervention and control groups as well. In the female control group, obesity prevalence increased markedly over the first years of elementary school and then decreased to less than 10% by the end of elementary school. By contrast, in the female intervention group, obesity prevalence decreased to reach 0% about midway through elementary school (see Fig. 7). The patterns of change for overweight status, as well as obesity and overweight status combined, were also quite similar between the female interventions versus control scenarios. However, the changes on overweight and obesity prevalence with the intervention are relevantly stronger on the low physically active groups in both genders (see Figs. 8 and 9). The initial analyses presented here indicate that the active furniture intervention in the elementary school environment has shown some impact on childhood obesity, although the effect is small and not statistically significant. Statistical tests on the impact of interventions are presented in Table V. This table presents the t-statistic and p-value for the difference on the means of the percentage of obese and overweight children with and without the intervention (dynamic furniture at school). The tests are presented for boys and girls and for each level of PA. This table presents that the p-value is relatively high for all cases (p-value $> = 0.314$). If we consider 90% confidence level ($\alpha = 10\%$), these results imply that there is no statistical difference on the mean percentage of obese and overweight children with and without the intervention for any level of PA.

Total child obesity and overweight prevalence in the U.S. is currently estimated as greater than 40% [20]. As of 2016, obesity prevalence among boys aged between 6 and 11 was 20.4% and had been increasing over the previous decade. At this time, obesity prevalence among girls was 16.3% and had held fairly steady for the previous decade [25], [51]. The model simulations

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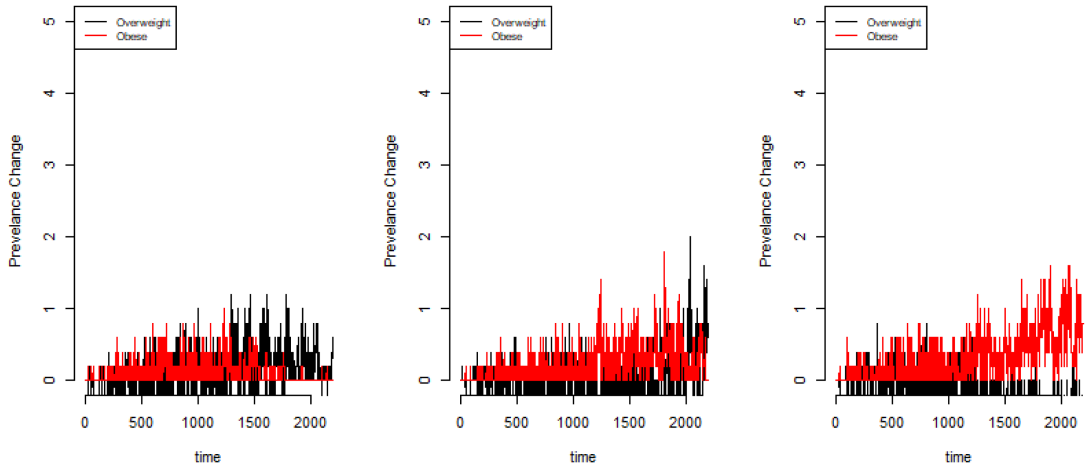

Fig. 9. Changes on girls overweight and obesity prevalence (%) (High, medium, and low activity levels, respectively) with the intervention in place.

TABLE V
STATISTICAL EVALUATION OF THE INTERVENTION BASED ON THE ACTIVITY GROUPS

Statistical Test of the Intervention Effect in Boys (No Intervention - Intervention)						
Physical Activity Level	High		Medium		Low	
	t-value	p-value	t-value	p-value	t-value	p-value
Obese	0.256	0.797	0.580	0.559	1.00	0.314
Overweight	0.530	0.595	0.008	0.931	-0.749	0.453
Statistical Test of the Intervention Effect in Girls (No Intervention - Intervention)						
Physical Activity Level	High		Medium		Low	
	t-value	p-value	t-value	p-value	t-value	p-value
Obesity Prevalence	0.211	0.8325	0.347	0.727	0.923	0.355
Overweight Prevalence	0.300	0.7635	0.670	0.502	0.415	0.677

produced initial average obesity and overweight prevalence that are higher than the national average (40.3% for males and 34.2% for females).

The average initial obesity prevalence for girls in the model was 15.9%, slightly higher than the 15.0% national statistic. However, the average initial obesity prevalence for boys in the model was 31.6%, higher than the 18.6% national statistic. In addition, the obesity trends in the modeled control (no intervention) scenarios do not appear to follow recent national trends of increasing obesity prevalence among boys and steady obesity prevalence among girls. Therefore, there may be some limitations in the base model that could be addressed with set-up rule modifications and model calibration prior to further testing of the intervention scenario. However, the logic of the intervention scenario calculations seems to be robust based on the current literature. For both males and females, higher PA profiles produced a marked reduction over time on overweight and obesity prevalence versus lower profiles.

Among males, regardless of PA profile, there were no apparent impacts of school dynamic furniture use on obesity and overweight prevalence trends. Among females with medium and high PA profiles, there were also no apparent impacts of school

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dynamic furniture use on obesity and overweight prevalence trends. Among females with a low PA profile, however, the overweight/obese prevalence trend lines began to separate at approximately second year with a slight prevalence reduction in the invention scenario versus the control scenario. At the sixth year, obese and overweight prevalence was 1% lower in the intervention group versus the control scenario. Although this result was not found statistically significant, it points to the higher positive effect of the intervention in the lower PA groups. This result suggests that by using expended energy acquisition devices, such as pedometers, more evidence can be collected in an experimental case study.

While the active furniture use does not present a statistically significant impact on the obesity-related health outcomes for high and medium activity groups, marginal effects are observed in the low physically active group, although this result was not statistically significant as well. However, the results may be sensitive in several parameters, e.g., age group and initial distribution of obesity prevalence. Regardless, the presented framework and the ABM can be used to investigate other PA-based interventions in collaboration with the active furniture use.

V. Conclusion

Given the high interest in public health interventions that aim to modify the school environment to address childhood obesity, this research set out to build an agent-based model that could be used to test such interventions in silico. We specifically sought to examine the potential impact of a dynamic furniture intervention on the prevalence of obesity over six years in elementary-school-aged children, holding other environmental variables constant. Such an investigation was of interest methodologically to overcome the inherent time and resource obstacles of longer term natural experiments, as well as the challenges of sampling and isolating environmental intervention variables in experiments.

While the intervention produced marginal changes in the prevalence trends over the considered time frame, dynamic furniture can be considered as a part of the broader systems wide interventions designed for preventing childhood obesity. Given the significance of built environment on health outcomes, especially on child health, any intervention presenting a positive impact on children's health should be considered in the complex ecosystem of public health interventions. Thus, the use of dynamic furniture in schools may be a worthwhile component of PA-oriented interventions, especially given its other known benefits, such as improved attention and learning [12]. In this article, we presented an approach to guide the use of an agent-based simulation models for assessing built-environment-based public health interventions. The agent-based modeling can be a resource-efficient method of exploring the potential efficacy of childhood obesity interventions, prior to committing resources to large intervention studies. When such studies do occur, the model could be formally validated based upon the empirical data.

The most obvious limitation of this research was that the experiments were simulated and did not occur in the real world with human subjects. However, the aim of this research was not to mimic actual population trends but rather to test the impact of one very specific environmental intervention on a school population, and the model drew upon established empirically derived parameters and formulas from the literature. That said, knowledge continues to grow as to the complexities and nuances of accurately modeling energy balance and growth in children. These recent conceptions may well displace older, simpler linear models of energy balance, and such work will no doubt contribute to the sophistication of further simulation model development and approaches for scenario testing in the future.

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Jeri Brittin received the Ph.D. degree in public health from the College of Public Health, University of Nebraska Medical Center, Omaha, NE, USA, in 2015.

She is the Director of Research at the global architecture and engineering firm HDR, she leads an award-winning, transdisciplinary team of researchers whose mandate is, very simply, to make the world better for people through planning and design. She trained as a Public Health Research Scientist in the realm of social and behavioral health promotion, and has a design and business background encompassing senior-level strategic and client service roles. Her research portfolio has focused on health and behavioral outcomes in healthcare settings, learning environments, workplaces, and community settings, including influences of transportation systems on health. She designs and conducts rigorous and actionable studies for clients across multiple industry and government sectors, and is well-versed in advanced statistical methods, qualitative and ethnographic research, and perspectives and approaches from systems science. Her work includes consumer studies and needs assessments to inform strategic and master planning, as well as predictive models and empirical evaluations of the impacts of designed interventions at a range of scales on human outcomes.

Dr. Brittin serves on several national committees focused on design and health.

Ozgur M. Araz received the Ph.D. degree in industrial engineering from Arizona State University, Tempe, AZ, USA, in 2009.

He is an Associate Professor with Supply Chain Management and Analytics Department, University of Nebraska-Lincoln (UNL), Lincoln, NE, USA. His main research areas are systems simulation, business analytics, healthcare operations, and public health informatics. Before joining the College of Business, UNL, he was a faculty member with the College of Public Health, University of Nebraska Medical Center. He completed his postdoctoral research fellowship with the Center for Computational Biology and Bioinformatics, University of Texas at Austin, Austin, TX, USA. His research appeared in journals, such as *Decision Sciences, Production and Operations Management, INFORMS Journal on Computing, Decision Support Systems, International Journal of Production Economics, Annals of Operations, Emerging Infectious Diseases, Medical Care,* and *Healthcare Management Science*, among many other prestigious journals.

Dr. Araz was an Associate Editor for *Decision Sciences*, *Transportation Research Part-E*, and *IISE Transactions of Healthcare Systems Engineering*, and an Area Editor for *Public Health Informatics in Health Systems Journal*.

Adrian Ramirez-Nafarrate received the Ph.D. degree in industrial engineering from Arizona State University, Tempe, AZ, USA, 2011.

He is a Professor with the School of Business, Universidad Panamericana, Guadalajara, Mexico. His research interests include simulation and optimization of manufacturing, logistics, and service systems. His research has appeared in journals, such as *Decision Sciences*, *Production and Operations Management*, *European Journal of Operational Research*, and *International Journal of Production Economics*, among many other prestigious journals. His research has been funded by federal and state science agencies in Mexico.

Terry T.-K. Huang received the B.A. degree in psychology from McGill University, Montreal, QC, Canada, in 1998, the MPH degree and the Ph.D. degree in preventive medicine both from the University of Southern California, Los Angeles, CA, USA, 2001, and the MBA degree from IE Business School, Madrid, Spain, in 2018.

He is a Professor and the Chair of Health Policy and Management Department and the Director of the Center for Systems and Community Design, City University of New York Graduate School of Public Health and Health Policy, New York, NY, USA. Prior to returning to academia, he played a leading role with the U.S. National Institutes of Health on the integration of systems science and chronic disease prevention. He has had a long history of research and policy leadership in the area of obesity. In addition, he is passionate about systems-oriented community health, design for health, social entrepreneurship, and strategies for collective impact. He has lectured and authored or coauthored extensively on these and other topics. Much of his current work focuses on innovations at the intersection of health, design, and business.

Dr. Huang was a recipient of the U.S. Department of Health and Human Services Secretary's Innovation Award in 2010, the NIH Director's Award in 2011, and the National Cancer Institute Award of Merit in 2012 and was named UNMC Distinguished Scientist in 2013. He is Board Certified in Public Health and Fellow, Councilor, and Past Program Chair of the Obesity Society. He is also VP North America of the World Obesity Federation.

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